

Project 3: SQL Data Analysis

Goal:

Use SQL queries to extract insights from a dataset.

Key Requirements:

- Write SELECT queries
- Use WHERE, ORDER BY, GROUP BY
- Perform basic aggregations (COUNT, SUM, AVG)

Key Skills:

SQL fundamentals, querying data, filtering and grouping



SQL File 12* x SQL File 13* SQL File 4* SQL File 6* SQL File 7* SQL File 8* SQL File 9*

Limit to 1000 rows

```
1 • CREATE TABLE Orders (  
2     OrderID VARCHAR(20),  
3     Date DATE,  
4     CustomerID VARCHAR(20),  
5     Product VARCHAR(50),  
6     Quantity INT,  
7     UnitPrice DECIMAL(10,2),  
8     ShippingAddress VARCHAR(100),  
9     PaymentMethod VARCHAR(50),  
10    OrderStatus VARCHAR(50),  
11    TrackingNumber VARCHAR(50),  
12    ItemsInCart INT,  
13    CouponCode VARCHAR(20),  
14    ReferralSource VARCHAR(50),  
15    TotalPrice DECIMAL(10,2)  
16 );
```

SQL File 12 SQL File 13 x SQL File 4 SQL File 6 SQL File 7 SQL File 8 SQL File 9

Limit to 1000 rows

```
1 • LOAD DATA INFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/ordersdatas.csv'
2 INTO TABLE Orders
3 CHARACTER SET latin1
4 FIELDS TERMINATED BY ','
5 LINES TERMINATED BY '\n'
6 IGNORE 1 ROWS
7 (OrderID, Date, CustomerID, Product, Quantity, UnitPrice,
8 ShippingAddress, PaymentMethod, OrderStatus,
9 TrackingNumber, ItemsInCart, CouponCode,
10 ReferralSource, TotalPrice);
11
```

SQL File 12 SQL File 13 SQL File 4 SQL File 6 SQL File 7 SQL File 8 SQL File 9

Limit to 1000 rows

```

1 • select * from ordersdata;
2
3 • SELECT OrderID, Product, Quantity, TotalPrice
4 FROM ordersdata;
5
6
7 • SELECT *
8 FROM ordersdata
9 WHERE TotalPrice > 2000 and OrderStatus = 'Delivered'
10 and PaymentMethod = 'Credit Card';
11

```

Result Grid Filter Rows: Export: Wrap Cell Content:

UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	CouponCode	ReferralSource	TotalPrice
33.26	895 Main St	Credit Card	Delivered	TRK37531040	9	SAVE10	Google	2166.3
59.85	197 Main St	Credit Card	Delivered	TRK32743531	9	SAVE10	Facebook	3299.25
56.07	587 Main St	Credit Card	Delivered	TRK41345634	6	SAVE10	Instagram	2830.35
77.82	811 Main St	Credit Card	Delivered	TRK47613018	0		Google	2311.28

ordersdata 13 ordersdata 14 ordersdata 15 x

SQL File 12*SQL File 13*SQL File 4*SQL File 6* xSQL File 7*SQL File 8*SQL File 9*



Limit to 1000 rows



1 • SELECT OrderID, Product, TotalPrice

2 FROM ordersdata

3 ORDER BY TotalPrice DESC;

4

5 • SELECT OrderID, Product, TotalPrice

6 FROM ordersdata

7 ORDER BY TotalPrice ASC;

Result Grid

	OrderID	Product	TotalPrice
▶	ORD201161	Phone	11.39
	ORD200863	Phone	14.06
	ORD200240	Tablet	17.24
	ORD200542	Tablet	17.98
	ORD2000000	Phone	10.00

Filter Rows:

Export: 

Wrap Cell Content: 

Fetch rows: 

ordersdata 2ordersdata 3 x

Output

SQL File 12* SQL File 13* SQL File 4* SQL File 6* SQL File 7* x SQL File 8* SQL File 9*

        Limit to 1000 rows     

```
1 • SELECT COUNT(*) AS TotalOrders,  
2 SUM(TotalPrice) AS TotalSales,  
3 AVG(TotalPrice) AS AverageOrderValue  
4 FROM ordersdata;
```

Result Grid   Filter Rows: Export:  Wrap Cell Content: 

	TotalOrders	TotalSales	AverageOrderValue
▶	1200	1264761.9600000004	1053.9683000000005

```
5
6 • SELECT PaymentMethod,
7       AVG(TotalPrice) AS AvgSales
8 FROM ordersdata
9 GROUP BY PaymentMethod;
10
11 • SELECT OrderStatus,
12       COUNT(*) AS NumberOfOrders
13 FROM ordersdata
14 GROUP BY OrderStatus;
```

	OrderStatus	NumberOfOrders
▶	Shipped	235
	Cancelled	250
	Returned	247
	Delivered	231



```
8          SUM(TotalPrice) AS Revenue
9  FROM ordersdata
10 GROUP BY ReferralSource
11 ORDER BY Revenue DESC;
12
13 • SELECT PaymentMethod,
14          SUM(TotalPrice) AS TotalRevenue
15  FROM ordersdata
16 GROUP BY PaymentMethod
17 ORDER BY TotalRevenue DESC;
```

Result Grid



Filter Rows:

Export:



Wrap Cell Content:



	PaymentMethod	TotalRevenue
▶	Credit Card	263847.62999999999
	Online	262442.94000000001
	Cash	259786.28999999992
	Gift Card	246323.92000000001
	...	...

Result 4

Result 5

Result 6 x

Output